Standard Operating Procedures Manual

Wastewater Treatment Plant: Liquid, Sludge and Biogas Systems

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Approval

Role	Name	Designation	Signature	Date
Prepared by		Process Engineer		
Reviewed by		HSE Manager		
Reviewed by		Maintenance Manager		
Approved by		Plant Manager		

1. Purpose

This manual defines the standard methods for operating the plant safely, responding to abnormal conditions, and keeping effluent within permit limits. It covers the safety procedures that apply plant-wide and the process procedures for the equipment shown in the plant flow diagram.

2. Scope

The manual applies to all operators, shift supervisors, maintenance technicians, laboratory staff and contractors working on the following process lines:

- **Liquid line:** sewers, lift station and wet well, inlet works (screens and grit), flow measurement, primary clarifiers, aeration tanks, secondary clarifiers, tertiary filtration, disinfection, effluent discharge or reuse.
- **Return flow:** RAS pumps from the secondary clarifiers back to the aeration tanks.
- **Sludge line:** primary sludge and waste activated sludge (WAS), thickening, anaerobic digestion, dewatering, cake disposal.
- **Gas line:** digester biogas, gas holder, CHP engine or boiler, flare.
- **Recycle:** centrate and filtrate returning to the plant inlet.

3. Important Notice on Numerical Values

All set points, alarm levels and operating ranges in this manual are **typical values for a mid-size plant**. Before this manual is issued for use, each value must be verified against the equipment manufacturer's data, the plant's design basis, the discharge permit, and applicable local regulations, and then signed off at the approval stage above.

4. Abbreviations

Term	Meaning
CHP	Combined heat and power (gas engine generator)
DO	Dissolved oxygen
DS	Dry solids
F/M	Food-to-microorganism ratio
HMI / SCADA	Operator screen / supervisory control system
LEL	Lower explosive limit
LOTO	Lockout / tagout
MLSS	Mixed liquor suspended solids
PPE	Personal protective equipment
PRV	Pressure relief valve
PTW	Permit to work

Term	Meaning
RAS	Return activated sludge
SRT	Sludge retention time (sludge age)
TSS	Total suspended solids
WAS	Waste activated sludge

5. SOP Register

SOP No.	Title	Process Area	Risk Level
SOP-GEN-001	Lockout / Tagout	All areas	High
SOP-GEN-002	Confined Space Entry	Wet well, tanks, digester	High
SOP-GEN-003	Alarm Response and Shift Handover	Control room	Medium
SOP-GEN-004	Incident Reporting and Regulatory Notification	All areas	Medium
SOP-GEN-005	Permit to Work and Hot Work Control	All areas, gas area	High
SOP-INL-001	Bar Screen Operation and Blockage Response	Inlet works	Medium
SOP-LFT-001	Lift Station High-High Level and Pump Clog Response	Lift station	High
SOP-BIO-001	Activated Sludge Foam Control	Aeration tanks	Medium
SOP-BIO-002	RAS Pump Failure and Sludge Blanket Control	Secondary clarifiers, RAS	High
SOP-GAS-001	Biogas Holder High Pressure, CHP Trip and Flare Operation	Gas line	High

SOP No.	Title	Process Area	Risk Level
SOP-SLD-001	Dewatering Centrifuge Operation and Centrate Recycle Control	Sludge line	Medium
SOP-LAB-001	Process Sampling and Monitoring Schedule	Laboratory	Low

6. Alarm Priority Convention

Priority	Meaning	Acknowledge Within	Operator on Location Within
P1	Risk to life, environment or permit compliance now	2 minutes	15 minutes
P2	Equipment or process degrading, likely to reach P1	5 minutes	30 minutes
P3	Advisory, no immediate effect	30 minutes	Next round

Part A. General Safety and Management SOPs

SOP-GEN-001: Lockout / Tagout (LOTO)

SOP No.	SOP-GEN-001	Revision	0
Process Area	All areas	Owner	HSE Manager

1. Purpose

To prevent unexpected start-up or release of stored energy while equipment is being cleaned, cleared, inspected or repaired.

2. Scope

All pumps, screens, scrapers, blowers, mixers, centrifuges, valves and electrical panels. This SOP must be followed **before** anyone places a hand, tool or body part inside or near moving or pressurised equipment.

3. Responsibilities

Role	Responsibility
Authorised person	Applies own lock and tag, performs the work
Shift supervisor	Authorises isolation, keeps the LOTO register
Qualified electrician	Performs electrical isolation and verification
HSE Manager	Audits LOTO records monthly

4. Required PPE

Safety shoes, gloves suited to the task, safety glasses, hard hat, and arc-rated clothing where electrical panels are opened.

5. Procedure

1. Identify the equipment and all its energy sources: electrical, hydraulic, pneumatic, gravity, trapped pressure, rotating inertia and process fluid.
2. Inform the shift supervisor and the control room. Confirm the process can run without the equipment, or arrange a standby.
3. Stop the equipment from the HMI or local station in the normal way.
4. Open the electrical isolator and, where fitted, close and lock process isolation valves.
5. Attach a personal lock and a tag to every isolation point. The tag shows name, date, time and reason. One person, one lock.
6. Release stored energy: vent pressure, drain liquid, block elevated parts, wait for rotation to stop.
7. **Verify zero energy:** attempt a start from the local station and the HMI. The equipment must not move. Return the switch to the off position.
8. Carry out the work. Do not remove or bypass any lock.
9. On completion, remove tools, refit guards and covers, and confirm everyone is clear.

10. Remove locks and tags, each by the person who applied it, and record the removal in the LOTO register.
11. Inform the control room, then restart the equipment under observation.

6. Special Rules

- A lock may be removed by someone else only under the supervisor's written lost-key procedure after confirming the owner is off site and the equipment is safe.
- Contractors apply their own locks alongside plant locks.

7. Records

LOTO register, completed isolation certificates (retain 3 years).

SOP-GEN-002: Confined Space Entry

SOP No.	SOP-GEN-002	Revision	0
Process Area	Wet well, tanks, sumps, digester	Owner	HSE Manager

1. Purpose

To control entry into spaces where toxic gas, low oxygen, flammable gas or drowning may be present.

2. Scope

Wet wells, clarifier and aeration tank sumps, channels deeper than 1.2 m, digesters, gas holders, thickener and sludge tanks. **Do not enter for routine clearing of jams. Use hoists, long-handled tools and suction equipment instead.**

3. Responsibilities

Role	Responsibility
Entry supervisor	Issues the entry permit, confirms rescue arrangements

Role	Responsibility
Entrant	Follows permit, leaves immediately on any alarm or discomfort
Standby attendant	Stays outside, keeps contact, never enters, summons rescue
Gas tester	Tests the atmosphere and records readings

4. Required Equipment

Calibrated 4-gas monitor (O_2, LEL, H_2S, CO), continuous personal monitors, forced ventilation, full-body harness with tripod and retrieval winch, communication device, escape breathing apparatus, rescue equipment.

5. Acceptable Atmosphere Limits

Parameter	Acceptable	Leave Immediately If
Oxygen	19.5 to 23.5 %	Below 19.5 % or above 23.5 %
Flammable gas	Below 10 % LEL	10 % LEL or above
Hydrogen sulfide (H_2S)	Below 10 ppm	10 ppm or above
Carbon monoxide (CO)	Below 25 ppm	25 ppm or above

6. Procedure

1. Confirm that entry is truly necessary and no alternative exists.
2. Isolate the space from inflow, outflow and energy sources under SOP-GEN-001. Isolate or blank connected pipework.
3. Complete the entry permit with the work, duration, entrants, attendant and rescue plan.
4. Ventilate the space for at least 15 minutes before testing.
5. Test the atmosphere at top, middle and bottom. Record the readings on the permit.
6. Brief all persons on hazards, communication signals and the emergency plan.
7. Entrants wear harness connected to the retrieval line. Continuous gas monitoring and ventilation stay on throughout.

8. The attendant keeps a head count and maintains contact. Re-test the atmosphere at least every 2 hours and after any break.
9. On completion, count out all persons and equipment, close the space, and close the permit.

7. Emergency

The attendant raises the alarm and starts non-entry retrieval. **No one enters to rescue unless trained and equipped as a rescue team.** Call the emergency services.

8. Records

Entry permit, gas test log, equipment calibration records (retain 3 years).

SOP-GEN-003: Alarm Response and Shift Handover

SOP No.	SOP-GEN-003	Revision	0
Process Area	Control room	Owner	Operations Manager

1. Purpose

To make sure every alarm is acknowledged, assessed and acted on consistently, and that no information is lost between shifts.

2. Procedure: Alarm Response

1. Acknowledge the alarm within the time in the Alarm Priority Convention (Section 6 of this manual).
2. Read the tag name, value, limit and time. Check related instruments to confirm the alarm is genuine and not a sensor fault.
3. Classify the alarm P1, P2 or P3.
4. For P1 and P2: inform the shift supervisor, then go to the location with a torch, gas monitor and radio.
5. Take the immediate safe action in the relevant process SOP.
6. If the value keeps moving toward the next limit, escalate to the supervisor and on-call maintenance.

7. Never disable, shelve or inhibit an alarm without the supervisor's written approval.
8. Record the alarm, action, outcome and time in the shift log.

3. Procedure: Shift Handover

1. The outgoing operator completes the shift log before handover.
2. Handover takes place face to face at the control panel and lasts at least 15 minutes.
3. The outgoing operator reports: open alarms, equipment out of service, active permits and LOTO, process changes, abnormal readings, chemicals and polymer stock, pending laboratory results.
4. The incoming operator reviews the log, walks the key areas, and signs to accept the plant.
5. The outgoing operator remains available until the incoming operator confirms understanding.

4. Records

Shift log, alarm history report (retain 3 years).

SOP-GEN-004: Incident Reporting and Regulatory Notification

SOP No.	SOP-GEN-004	Revision	0
Process Area	All areas	Owner	Plant Manager

1. Purpose

To ensure every incident, near miss and permit exceedance is reported, investigated and prevented from recurring.

2. Procedure

1. Make the situation safe first. Protect people, then the environment, then the equipment.
2. Notify the shift supervisor immediately. The supervisor notifies the Plant Manager for all P1 events.
3. Preserve evidence: do not clean or repair until photos are taken and readings recorded.

4. Complete the incident report (Appendix B) within 24 hours.
5. **Regulatory notification:** If an overflow, bypass, permit exceedance or gas release has occurred, the Plant Manager notifies the regulator within the time set in the discharge permit and applicable law.
6. Carry out a root-cause investigation within 7 days for P1 events and within 14 days for P2 events.
7. Raise corrective and preventive actions with owner and due date. Track to closure.
8. Share lessons learned at the next safety meeting and update the affected SOPs.

3. Records

Incident report, investigation report, action tracker (retain 5 years).

Part B. Process SOPs

SOP-INL-001: Bar Screen Operation and Blockage Response

SOP No.	SOP-INL-001	Revision	0
Process Area	Inlet works	Owner	Operations Manager

1. Purpose

To keep the mechanical bar screens removing rags and debris reliably, and to respond safely when a screen blocks.

2. Scope

Mechanical bar screens BS-1 and BS-2, the manual bypass bar rack and the screenings conveyor.

3. Responsibilities

Role	Responsibility
Night / shift operator	Monitors, responds to alarms, clears blockages after isolation

Role	Responsibility
Shift supervisor	Approves bypass use, informs the lift station team
Maintenance technician	Repairs rake, motor and drive faults

4. Required PPE

Gloves, safety glasses, face shield, rubber boots, coverall, hard hat. Hand hygiene after the task. Up-to-date hepatitis and tetanus vaccination.

5. Operating Limits

Parameter	Normal	Alarm / Limit	Action
Differential level across screen	5 to 15 cm	30 cm (high)	Start this SOP, Section 7
Upstream channel level	0.8 to 1.2 m	1.5 m action · 1.8 m overflow weir	Open bypass at 1.5 m
Rake motor current	6 to 9 A	14 A (overload trip)	Isolate before inspection
Inlet flow per screen	900 to 1,200 m³/h (plant)	1,500 m³/h (single-screen capacity)	Request flow reduction

6. Routine Operation

1. Before forecast heavy rain, confirm **both** screens are available. Do not schedule screen maintenance during storm warnings.
2. Each shift, walk the inlet works. Check rake movement, motor sound, wash water, and conveyor.
3. Empty the screenings container before it is three-quarters full.
4. Record differential level and screenings volume each shift.

7. Procedure: Blockage Response (High Differential Level)

1. Acknowledge the alarm and inform the supervisor. Check the second screen’s status.

2. If only one screen is available, ask the lift station operator to reduce pumping to one pump so flow stays below 1,500 m³/h.
3. Go to the screen. Check the differential level, upstream channel level and motor condition. Note whether the overload has tripped.
4. If the upstream channel level reaches **1.5 m**, open the manual bypass to the coarse bar rack. Staff the rack with a second person.
5. **Isolate the jammed screen under SOP-GEN-001. Never reach into or onto a rake that is not locked out.**
6. Clear the jam from the walkway with a long-handled hook. Remove rags in pieces and place them in the screenings container.
7. Inspect the rake, chain, and shear pin for damage. Record any defects for repair.
8. Remove locks. Restart and observe at least three full rake cycles. Motor current must return to 6 to 9 A.
9. Confirm the channel level falls below 1.2 m and the differential level below 15 cm.
10. Close the bypass. Remove and dispose of material trapped on the bypass rack.
11. Inspect the grit chamber and downstream pumps for carried-over rags during the next two rounds.
12. Record the event, times, levels, and parts used in the shift log.

8. Escalation

If the channel level reaches 1.8 m (overflow weir), the event becomes P1. Notify the Plant Manager and follow SOP-GEN-004.

9. Records

Shift log, screenings log, maintenance work order (retain 3 years).

SOP-LFT-001: Lift Station High-High Level and Pump Clog Response

SOP No.	SOP-LFT-001	Revision	0
Process Area	Lift station and wet well	Owner	Maintenance Manager

1. Purpose

To prevent sewer overflow and flooding when the lift station pumps clog, trip or fail.

2. Scope

Submersible pumps P-101 and P-102, wet well level instruments, davit lifting equipment, mobile pump connection, and the vacuum tanker arrangement.

3. Responsibilities

Role	Responsibility
On-call technician	Attends, clears the pump, restarts
Shift supervisor	Authorises tanker call-out and external notification
Control room operator	Monitors level remotely, adjusts the inflow where possible

4. Required PPE

High-visibility vest, gloves, safety shoes, hard hat, personal H_2S monitor. Traffic cones for roadside stations.

5. Operating Limits

Parameter	Normal	Alarm / Limit	Action
Wet well level	1.0 to 2.4 m	3.2 m high · 3.6 m high-high · 4.0 m spill	Dispatch at 3.2 m; tanker at 3.6 m
Pump motor current	28 to 34 A	42 A overload · 24 A low-load	Investigate
Pump winding temperature	55 to 75 °C	105 °C trip	Do not restart until cool
Pump discharge pressure	2.8 to 3.4 bar	2.0 bar low	Suspect clog
Pump flow	380 to 450 m³/h	300 m³/h low	Suspect clog

6. Procedure

1. Acknowledge the HIGH level alarm (3.2 m). Check both pumps' status, current and pressure on SCADA.
2. Dispatch the on-call technician. Target arrival within 30 minutes.
3. On arrival, park clear of the hatch, place cones, and test the atmosphere with the personal monitor before opening the cover. **Do not enter the wet well.** Use SOP-GEN-002 if entry is unavoidable.
4. Check the pump status, motor temperature and local level reading. If level is at or above **3.6 m (high-high)**, request the vacuum tanker immediately and connect the mobile pump if available.
5. Reduce the inflow where the network allows it, for example by closing a gate to a non-critical branch under supervisor approval.
6. Isolate the failed pump under SOP-GEN-001. Hoist it using the davit crane. Do not lift by the power cable.
7. Remove the debris from the impeller and volute. Check the impeller for wear, the seal for leaks, and the cable for damage.
8. Allow the motor to cool to normal winding temperature (below 75 °C) and check insulation resistance before reset. Identify the cause **before** resetting the thermal overload. Do not reset repeatedly.

9. Lower the pump, restore power, and start it in hand mode. Verify motor current 28 to 34 A, discharge pressure 2.8 to 3.4 bar, flow 380 m³/h or above.
10. Repeat the process for the second pump if needed. Return both pumps to automatic duty and standby rotation.
11. Watch the wet well level fall below 2.4 m. Release the tanker.
12. Warn the inlet works operator about the surge of flow expected as the level is pumped down.

7. Escalation

If the level reaches 3.85 m or a spill is imminent, the event is P1. Notify the Plant Manager. If sewage has escaped, follow SOP-GEN-004 for regulatory notification.

8. Preventive Actions

Monthly pump rotation and run test, quarterly wet well cleaning, quarterly inspection of impellers, test of level float switches and mobile pump connection twice a year.

9. Records

Shift log, pump maintenance record, tanker call-out record, incident report where applicable (retain 5 years).

SOP-GEN-005: Permit to Work and Hot Work Control

SOP No.	SOP-GEN-005	Revision	0
Process Area	All areas, gas area	Owner	HSE Manager

1. Purpose

To control non-routine and high-risk work, especially any work that can create a spark or flame near biogas.

2. Scope

Hot work (welding, cutting, grinding, open flame), work in the classified gas area, electrical work on live panels, work at height, lifting operations and excavation.

3. Procedure

1. The requester submits a permit request describing the work, location, tools, people and duration.
2. The issuer inspects the site with the requester and confirms the hazards and controls.
3. For hot work: remove or cover combustible material, test for flammable gas immediately before work starts (must read **0 % LEL**), and station a fire watch with an extinguisher.
4. **No hot work is permitted in the gas area or within the exclusion distance of the gas holder, digester, flare or gas pipework while biogas is being vented, flared at high rate, or while any gas alarm is active.**
5. Isolate energy under SOP-GEN-001 and complete confined space entry under SOP-GEN-002 where needed.
6. The permit is valid for one shift only. It is displayed at the work site and signed by the issuer, receiver and the control room.
7. At completion, the receiver inspects the area, confirms the plant is safe, and closes the permit. The fire watch remains for 60 minutes after hot work ends.
8. Suspended permits are cancelled and re-issued after any change in conditions or any gas alarm.

4. Records

Permits and gas test records (retain 3 years).

SOP-BIO-001: Activated Sludge Foam Control

SOP No.	SOP-BIO-001	Revision	0
Process Area	Aeration tanks	Owner	Process Engineer

1. Purpose

To detect, control and prevent stable biological foam (such as that caused by *Nocardia*) on aeration tanks and secondary clarifiers.

2. Scope

Aeration tanks, blowers and diffusers, surface scum trap, foam spray nozzles, WAS pumps, and the secondary clarifier weirs.

3. Responsibilities

Role	Responsibility
Process Engineer	Diagnoses cause, sets WAS rate and targets
Operator	Records foam depth, runs spray and skimming
Laboratory	Microscopic examination, MLSS, sludge volume index

4. Required PPE

Slip-resistant footwear, gloves, safety glasses, life jacket where required. Foam makes walkways slippery: barrier and sign the affected area.

5. Operating Limits

Parameter	Normal	Alarm / Limit	Target after Control
Sludge age (SRT)	5 to 10 days	15 days	About 11 days
WAS rate	180 to 250 m³/d	Below 120 m³/d	260 m³/d during recovery
MLSS	2,500 to 3,500 mg/L	4,500 mg/L	3,300 mg/L
F/M ratio	0.2 to 0.4	Below 0.15	About 0.28
Dissolved oxygen	1.5 to 3.0 mg/L	Below 1.0 mg/L	2.0 mg/L or above
Foam depth	Below 2 cm	10 cm	Below 5 cm
Effluent turbidity	2 to 5 NTU	10 NTU (permit)	Below 5 NTU

6. Procedure

1. Measure and record foam depth at least twice per shift. Flag any foam that is thick, brown and persistent.

2. Ask the laboratory for microscopic identification. Confirm the filament type before changing process control.
3. Calculate the current sludge age from MLSS, tank volume and WAS flow.
4. If sludge age exceeds target, **increase the WAS rate in steps.** Do not change sludge age by more than 25 % in one day, to avoid shocking the biology.
5. Verify DO stays at 1.5 mg/L or above after each change. Adjust blower output if needed.
6. Operate the foam spray nozzles and surface scum trap. Remove foam from the tank surface mechanically and keep the walkway clear.
7. **Do not pump trapped foam directly to the digester without approval.** It can cause foaming in the digester.
8. Check for grease sources: primary clarifier scum removal, trade discharges and any fat, oil and grease in the inflow. Report findings to the Process Engineer.
9. Use anti-foam chemicals only on Process Engineer approval, as a short-term measure.
10. Watch the secondary clarifier weirs and effluent turbidity. If effluent turbidity exceeds 8 NTU, increase sampling to hourly and follow SOP-BIO-002 clarifier checks.
11. Review progress daily until foam depth stays below 5 cm for 3 consecutive days.

7. Escalation

If effluent TSS reaches 25 mg/L or turbidity reaches 10 NTU, inform the Plant Manager. Follow SOP-GEN-004 if a permit limit is exceeded.

8. Records

Foam log, WAS rate record, microscopy report, laboratory results (retain 3 years).

SOP-BIO-002: RAS Pump Failure and Sludge Blanket Control

SOP No.	SOP-BIO-002	Revision	0
Process Area	Secondary clarifiers, RAS pumps	Owner	Operations Manager

1. Purpose

To keep the sludge blanket in the clarifier below overflow level and the aeration tank supplied with returned sludge when a RAS pump fails.

2. Scope

RAS pumps RAS-P1 and RAS-P2, the portable standby pump, the clarifier scraper bridge, blanket level probe, and WAS draw-off.

3. Responsibilities

Role	Responsibility
Night / shift operator	Responds, starts standby, monitors blanket
Electrician	Diagnoses motor and breaker fault
Process Engineer	Decides wasting and flow-reduction strategy

4. Required PPE

Gloves, safety shoes, hard hat, safety glasses, arc-rated PPE for panel work.

5. Operating Limits

Parameter	Normal	Alarm / Limit	Action
Total RAS flow	700 to 840 m^3/h	550 m^3/h low	Check pumps
RAS pump flow (each)	350 to 420 m^3/h	n/a	Start standby if one is lost
RAS pump winding temperature	60 to 80 °C	130 °C trip	Do not reset until cool
Sludge blanket depth	0.3 to 0.8 m	1.5 m high · about 2.5 m weir level	Act at 1.5 m
Clarifier solids loading	3 to 5 $kg/m^2/h$	6.5 $kg/m^2/h$	Reduce flow
Aeration MLSS	2,500 to 3,500 mg/L	2,000 mg/L low	Restore RAS
Effluent TSS	8 to 15 mg/L	30 mg/L (permit)	Sample hourly

6. Procedure

1. Acknowledge the SLUDGE BLANKET HIGH alarm. Check the actual blanket depth with the probe, then confirm with a manual sample.
2. Check the status of both RAS pumps on SCADA. Identify tripped breakers, motor temperature and local fault displays.
3. Run the remaining RAS pump at maximum permitted speed.
4. Start the portable standby pump and connect it to the RAS line. Confirm total RAS flow is climbing back toward 700 m³/h or above.
5. Ask the process engineer to approve a temporary reduction of plant inflow (for example, by setting the lift station to one pump) to cut clarifier solids loading.
6. If the blanket exceeds 2.0 m, increase WAS withdrawal temporarily to remove solids from the system, to the process engineer's instruction.
7. Call the electrician. Isolate the failed pump under SOP-GEN-001. Inspect the suction line for blockage, check the winding resistance and insulation, bearings and the coupling.
8. **Do not repeatedly reset a tripped breaker.** Reset only after the cause is identified.
9. Restart the repaired pump in hand mode and verify flow 350 to 420 m³/h. Return to automatic duty/standby.
10. Reduce the temporary flow limit gradually and watch the blanket fall below 0.8 m. This normally takes 4 to 8 hours.
11. Sample the effluent hourly until TSS is back below 15 mg/L.

7. Escalation

If the blanket reaches 2.5 m or effluent TSS reaches 30 mg/L, the event is P1. Notify the Plant Manager and follow SOP-GEN-004.

8. Records

Shift log, blanket depth readings, effluent results, pump repair report (retain 3 years).

SOP-GAS-001: Biogas Holder High Pressure, CHP Trip and Flare Operation

SOP No.	SOP-GAS-001	Revision	0
Process Area	Gas line	Owner	Maintenance Manager

1. Purpose

To manage biogas safely when the CHP engine or boiler cannot consume gas, so that holder pressure stays within limits and no uncontrolled gas release occurs.

2. Scope

Digester gas space, gas holder with pressure relief valve, gas pipework, CHP engine or boiler, emergency flare stack, and gas detection system.

3. Responsibilities

Role	Responsibility
Shift supervisor	Commands the response, controls the exclusion zone
Operator	Verifies flare, monitors pressure, checks gas detectors
CHP technician	Diagnoses and restarts the engine
HSE Manager	Reviews zone control and gas monitoring

4. Hazards

Biogas is mostly methane and is flammable and explosive; it also contains hydrogen sulfide, which is toxic. Ignition sources and uncontrolled releases are the main risks.

5. Required PPE and Equipment

Flame-resistant clothing, anti-static safety shoes, personal 4-gas monitor, intrinsically safe torch and radio in the gas area. Mobile phones not rated for the zone must stay outside it.

6. Operating Limits

Parameter	Normal	Alarm / Limit	Action
Holder pressure	15 to 25 mbar	32 mbar alarm · 38 mbar relief opens	Verify flare at 32 mbar
Holder fill level	40 to 70 %	90 % high	Verify flare, reduce digester feed
Biogas production	160 to 200 m³/h	n/a	Record
CHP gas consumption	150 to 190 m³/h	100 m³/h low	Investigate engine
CHP electrical output	380 to 420 kWe	0 kWe tripped	Call technician
Gas near holder vent	0 % LEL	10 % LEL	Evacuate zone, P1

7. Procedure: High Holder Pressure after CHP Trip

1. Acknowledge the siren and the GAS HOLDER PRESSURE HIGH alarm. Inform the shift supervisor.
2. Set up the **exclusion zone** around the gas area. Stop all hot work and cancel active permits under SOP-GEN-005. Allow no ignition sources.
3. Send the operator to the flare with a gas monitor. Verify that the flare valve has opened, the pilot or igniter is lit, and the flame detector shows a flame.
4. Check on SCADA that holder pressure is falling and the flare gas flow reads close to biogas production (about 185 m³/h).
5. If the flare has not started automatically, follow the flare manual start steps. Do not override a flame-failure trip and do not bypass any gas safety interlock.

6. Check the gas detectors near the holder, PRV vent and engine room. Any reading of **10 % LEL or above** is a P1 event: evacuate the zone, call the fire service, and inform the Plant Manager.
7. If the holder fill reaches 90 % and pressure stays above 32 mbar, reduce or stop sludge feed to the digester to lower gas production.
8. Call the CHP technician. Record the engine fault code and run hours. Do not restart the engine until the fault is understood and repaired.
9. Restart the engine following the manufacturer's start sequence. Verify gas supply pressure, engine load 380 to 420 kWe and gas consumption 150 to 190 m³/h.
10. Confirm the flare closes and holder pressure returns to 15 to 25 mbar.
11. Remove the exclusion zone and restore permits. Record the event, flare duration and estimated gas flared.

8. Escalation

If pressure reaches 38 mbar and the flare is not burning, or the relief valve is venting, the event is P1. Evacuate to the muster point, call emergency services, and follow SOP-GEN-004.

9. Records

Flare log, gas detector report, CHP fault report, shift log (retain 5 years).

SOP-SLD-001: Dewatering Centrifuge Operation and Centrate Recycle Control

SOP No.	SOP-SLD-001	Revision	0
Process Area	Sludge line	Owner	Process Engineer

1. Purpose

To operate the dewatering centrifuge so that cake dryness and solids capture stay in range, and to prevent ammonia-rich centrate from overloading the liquid treatment line when it returns to the plant inlet.

2. Scope

Dewatering centrifuge, polymer preparation and dosing system, sludge feed pumps, centrate line and centrate return to the plant inlet, cake conveyor and hopper.

3. Responsibilities

Role	Responsibility
Sludge area operator	Starts, monitors and adjusts the centrifuge
Maintenance technician	Repairs, resets setpoints to approved values after work
Process Engineer	Approves setpoints, manages centrate return strategy
Laboratory	Tests centrate, cake and plant inlet ammonia

4. Required PPE

Gloves, safety glasses, hearing protection, safety shoes, coverall. Polymer powder is slippery when wet; clean spills immediately.

5. Operating Limits

Parameter	Normal	Alarm / Limit	Action
Bowl speed	3,200 rpm	Below 2,900 rpm	Stop feed, check drive
Polymer dose	8 to 12 kg/t DS	Below 6 kg/t DS	Check dosing pump and concentration
Solids capture	92 to 97 %	88 %	Adjust, then re-test
Cake dryness	22 to 26 % DS	20 % DS	Adjust differential speed and polymer
Centrate ammonia (NH_4-N)	800 to 1,100 mg/L	1,400 mg/L	Control return rate
Plant inlet ammonia	35 to 42 mg/L	50 mg/L	Notify process engineer

Parameter	Normal	Alarm / Limit	Action
Blower airflow	2,800 to 3,400 Nm³/h	4,200 Nm³/h max	Reduce centrate return
Effluent ammonia	0.5 to 2 mg/L	5 mg/L (permit)	Escalate

6. Procedure: Start-up after Maintenance

1. Confirm all work permits and LOTO are closed (SOP-GEN-001 and SOP-GEN-005).
2. **Verify setpoints against the approved setpoint sheet** (bowl speed 3,200 rpm, differential speed, pond depth, polymer dose). The maintenance technician and operator sign the commissioning check sheet. No setpoints may be changed without Process Engineer approval.
3. Start the polymer system and confirm the concentration is in the approved range.
4. Start the centrifuge and bring it to full speed before feeding sludge.
5. Start the sludge feed at the lower rate and increase gradually.
6. After 30 minutes, take grab samples of feed, cake and centrate. The laboratory reports capture rate and cake dryness.
7. Adjust and re-sample until readings are in the Operating Limits table.

7. Procedure: Abnormal Centrate Quality

1. Acknowledge the alarm or laboratory report showing low capture, low cake dryness, or high centrate ammonia.
2. Check bowl speed, polymer dosing rate, feed solids and feed rate against the approved setpoints. Correct any deviation.
3. If the cause is a setting error (for example after repair), restore the approved setpoints and record the correction.
4. Notify the Process Engineer. Check **plant inlet ammonia** and blower airflow. A rise of more than 15 % over normal indicates the recycle stream is affecting the liquid line.
5. Slow the centrate return. Hold centrate in the buffer tank and **return it gradually over 24 hours, preferably off-peak**, instead of in a single slug.
6. Increase the frequency of effluent ammonia sampling to every 4 hours until inlet ammonia is back below 42 mg/L.

7. Where blower airflow approaches 4,200 Nm^3/h, pause centrifuge feed and further centrate return, with the Process Engineer's approval.
8. Record the event and corrective action. Add the cause to the incident log under SOP-GEN-004.

8. Records

Commissioning check sheet, setpoint log, centrate and cake results, inlet ammonia trend (retain 3 years).

SOP-LAB-001: Process Sampling and Monitoring Schedule

SOP No.	SOP-LAB-001	Revision	0
Process Area	Laboratory	Owner	Laboratory Manager

1. Purpose

To provide consistent process data for control decisions and compliance reporting.

2. Procedure

1. Collect samples from the locations and at the frequencies below, using clean, labelled containers.
2. Preserve and transport samples according to the laboratory method. Record the time and the sampler.
3. Report any result outside the limits in the process SOPs to the shift supervisor immediately.
4. Calibrate online instruments (pH, DO, turbidity, flow, gas detectors) at the interval set by the manufacturer. Record every calibration.

3. Schedule

Sample Point	Parameters	Frequency
Plant inlet (after grit)	Flow, BOD, COD, TSS, NH_4-N	Daily composite

Sample Point	Parameters	Frequency
Aeration tank	MLSS, DO, sludge volume index, microscopy	Daily (microscopy weekly)
RAS and WAS lines	Solids concentration	Daily
Secondary clarifier	Blanket depth	Every shift
Final effluent	BOD, TSS, NH_4-N, turbidity, pH, residual disinfectant	Daily (turbidity continuous)
Centrate and cake	NH_4-N, TSS, cake dryness	Daily and after any maintenance
Digester gas	Methane, H_2S	Weekly

4. Records

Laboratory register and results database (retain 5 years).

Appendices

Appendix A. Consolidated Limits at a Glance

Area	Parameter	Normal	Alarm / Limit
Inlet screens	Differential level	5 to 15 cm	30 cm
Inlet screens	Channel level	0.8 to 1.2 m	1.8 m (weir)
Lift station	Wet well level	1.0 to 2.4 m	3.6 m high-high · 4.0 m spill
Lift station	Pump pressure	2.8 to 3.4 bar	2.0 bar
Aeration	Sludge age	5 to 10 days	15 days
Aeration	Foam depth	Below 2 cm	10 cm
Clarifier	Blanket depth	0.3 to 0.8 m	1.5 m alarm · 2.5 m overflow
RAS	Total flow	700 to 840 m^3/h	550 m^3/h

Area	Parameter	Normal	Alarm / Limit
Gas holder	Pressure	15 to 25 mbar	32 mbar alarm · 38 mbar relief
Gas holder	Fill level	40 to 70 %	90 %
Centrifuge	Capture rate	92 to 97 %	88 %
Plant inlet	Ammonia	35 to 42 mg/L	50 mg/L
Effluent	Ammonia	0.5 to 2 mg/L	5 mg/L (permit)
Effluent	TSS	8 to 15 mg/L	30 mg/L (permit)

Appendix B. Incident Report Form

Field	Entry
Report No.	
Date and time of incident	
Location / equipment tag	
Reported by	
Priority (P1 / P2 / P3)	
Description of what happened	
Readings at the time (levels, pressures, flows, quality)	
Immediate actions taken	
People injured / environmental release	
Regulator notified (when, by whom)	
Root cause	
Corrective and preventive actions (owner, due date)	
SOPs to be updated	
Reviewed by / date	

Appendix C. Revision History

Revision	Date	Description of Change	Prepared by	Approved by
0	[DD-MMM-YYYY]	Initial issue for review		